## Lesson 4 Sets

In this lesson, we continue to build the foundation of rigorous math, focusing on sets, which are used extensively in math.

- A ***set*** is an **unordered** and **non-repetitious** collection of objects. An object $x$ in a set $A$ is an ***element*** of $A$, denoted by $x \in A$ ($x \notin A$ means $x$ is not in $A$). Two sets are equal, $A = B$, if $A$ and $B$ contain exactly the same elements.

The sets $A = \{1, 2\}$ and $B = \{2, 1\}$ are the same set; $1 \in A$ and $3 \notin A$. We define a set by listing the elements or using the **set-builder notation** {variable :attributes} or {form of elements :attributes}, where ":" means "such that".

From the set of integers $\mathbb{Z}$ and the set of real numbers $\mathbb{R}$, we define other number sets. The **positive integers** are often called **natural numbers**, denoted $\mathbb{N}$.

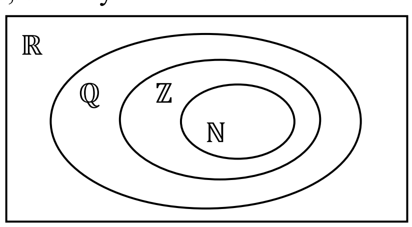
- $\mathbb{N}$ given as a list: $A = \{1, 2, 3, 4, \dots\}$
- $\mathbb{N}$ given by set-builder notation: $\mathbb{N} = \{x \in \mathbb{Z} : x > 0\}$

The set of **rational numbers** is $\mathbb{Q} = \{m/n : m, n \in \mathbb{Z}, n \neq 0\}$.

In a given context, we may define a **universal set** $U$, which includes all objects of interest. Within the universe, we can define other sets.

**Definition** Set $A$ is a **subset** of set $B$ $(A \subseteq B)$ if every element of $A$ is in $B$.

For example, we may take the universal set to be $\mathbb{R}$, then $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$.




A **Venn diagram** is a useful visual tool for sets.

Any set is a subset of the universal set. Another special set, the **empty set**, denoted $\emptyset$ or { }, has no elements and is a subset of every set. We use $\nsubseteq$ for "*not* a subset".

**Definition** The **power set** of a set $A$, denoted $\mathcal{P}(A)$, is the set of all subsets of $A$.

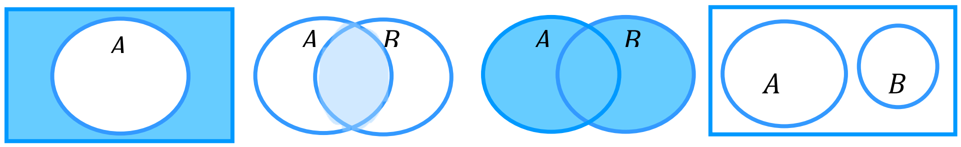
The power set of set $A = \{1, 2\}$ is $\mathcal{P}(A) = \big\{\emptyset, \{1\}, \{2\}, \{1, 2\}\big\}$, and we have $1 \in A$, $\{1\} \subseteq A$, $\{1\} \in \mathcal{P}(A)$, and $\big\{\{1\}\big\} \subseteq \mathcal{P}(A)$, but $\{1\} \nsubseteq \mathcal{P}(A)$. It is very important to distinguish between $\subseteq$ (subset) and $\in$ (element)!

Just as we have operations such as $+$ and $\times$ on numbers, and connectives such as $\wedge$ and $\vee$ on statements, we also have **set operations**, on sets.

**Definition** The **complement** of a set $A$ is $\overline{A} = \{x \in U : x \notin A\}$.
**Definition** The **intersection** of sets $A$ and $B$ is $A \cap B = \{x : x \in A \textbf{ and } x \in B\}$.
**Definition** The **union** of sets $A$ and $B$ is $A \cup B = \{x : x \in A \textbf{ or } x \in B\}$.

The union includes elements which are in both sets, as "or" is *inclusive* in logic. There is a parallel between **set operations** and **connectives**:

$$\boldsymbol{\sim p \leftrightarrow \overline{A}}, \qquad \boldsymbol{p \wedge q \leftrightarrow A \cap B}, \qquad \boldsymbol{p \vee q \leftrightarrow A \cup B}$$

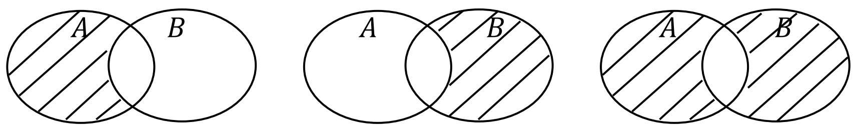



$\overline{A}$ $\quad$ $A \cap B$ $\quad$ $A \cup B$ $\quad$ Disjoint sets

**Definition** Two sets $A$ and $B$ are **disjoint** if $A \cap B = \emptyset$.

**Definition** Let $A$ and $B$ be sets.

- The **set difference** of $A$ and $B$ is $A \backslash B = \{x : x \in A \; and \; x \notin B\}$
- The **symmetric difference** of $A$ and $B$ is $A \, \Delta \, B = (A \backslash B) \cup (B \backslash A)$




$A \backslash B$ $\quad$ $B \backslash A$ $\quad$ $A \, \Delta \, B$

Let $U = \{1, 2, 3, \ldots, 9\}$ (universal set), $A = \{2, 4, 6, 8\}$, and $B = \{3, 6, 9\}$.

- $\overline{A} = \{1, 3, 5, 7, 9\}$, $A \cap B = \{6\}$, $A \cup B = \{2, 3, 4, 6, 8, 9\}$
- $A \backslash B = \{2, 4, 8\}$, $B \backslash A = \{3,9\}$, $A \, \Delta \, B = \{2, 3, 4, 8, 9\}$

Any set $A$ and its complement $\overline{A}$ are disjoint as there cannot be an element which is both in $A$ and not in $A$. By definition, $\overline{B} = U \backslash B$, so $A \backslash B$ is the complement of $B$ *relative to* $A$ or in $A$, while $\overline{B} = U \backslash B$ is the complement of $B$ in the universal set $U$.

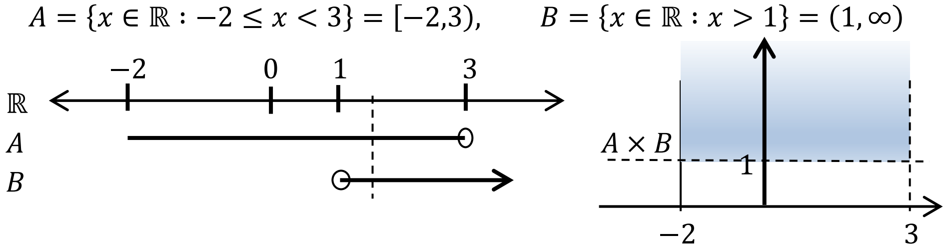
**Definition** Let $A$ and $B$ be sets. The **Cartesian product** of $A$ and $B$ is

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

Let $A = \{2, 4\}$ and $B = \{3, 6, 9\}$. then $A \times B = \{(2,3), (2, 6), (2, 9),$
$(4,3), (4, 6), (4, 9)\}$

The Cartesian $xy$-plane is the set of all coordinates $(x, y)$ or the Cartesian product $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y): x, y \in \mathbb{R}\}$. A subset of $\mathbb{R}$ is part of the real line, and a subset of $\mathbb{R}^2$ is part of the $xy$-plane. A continuous subset of $\mathbb{R}$ is called an **interval**. Let

$$A = \{x \in \mathbb{R} : -2 \leq x < 3\} = [-2,3), \qquad B = \{x \in \mathbb{R} : x > 1\} = (1, \infty)$$




The open circle and a round bracket mean that the endpoint is not included in the interval; a square bracket means that the endpoint is included. The interval $(a, b)$ is **open**, $[a, b]$ is **closed**, and $(a, b]$ and $[a, b)$ are neither open nor closed.